

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm. Examination

University Theory Examination March 2018

PH116 Pharmaceutical Biology-I

Date: 28.03.2018, Wednesday Time: 10:00 a.m. to 1:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
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SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. The structural area where two bones meet is known as a :
[A] Junction
[B] Joint
[C] Endocrine gland
[D] Lymph node
 2. Muscle action can be classified as:
[A] Sympathetic and parasympathetic
[B] Voluntary and involuntary
[C] Autonomic and sympathetic
[D] Autonomic and voluntary
 3. Which of these is not used by the body for energy?
[A] Fat
[B] Carbohydrate
[C] Protein
[D] Water
 4. Which cells produce the pigment that contributes to hair colour?
[A] Keratinocytes
[B] Melanocytes
[C] Langerhans cells
[D] Merkel cells
 5. During exercise, the greatest increase in blood flow occurs in:
[A] Skeletal muscle
[B] Cardiac muscle
[C] Kidneys
[D] None of the above

6. In which part of a cell does the process of making ATP from oxygen and glucose take place?
[A] Lysosomes
[B] Ribosomes
[C] Mitochondria
[D] Golgi apparatus
7. Which cell organelles contain an acidic environment capable of digesting a wide variety of molecules?
[A] Lysosomes
[B] Ribosomes
[C] Centrosomes
[D] Golgi complex
8. What are tendons and ligaments composed of?
[A] Dense connective tissue
[B] Liquid connective tissue
[C] Muscular tissue
[D] Epithelial tissue
9. Which of the following is NOT part of the plasma membrane of a cell?
[A] Integral proteins
[B] Glycoproteins
[C] Plasma proteins
[D] Peripheral proteins
10. The plasma membrane of a cell contains molecules that have a hydrophobic end and a hydrophilic end. What are they called?
[A] Phospholipids
[B] Cholesterol
[C] Integral proteins
[D] Glycoproteins
11. The perception of color comes from all the following except:
[A] Red cones
[B] Green cones
[C] Blue cones
[D] Yellow cones
12. Following is a contracting protein.
[A] Tropomyosin
[B] Troponin
[C] Myosin
[D] Titin
13. _____ tissue covers body surfaces and lines hollow organs, body cavities, and ducts.
[A] Nervous Tissue
[B] Muscular Tissue
[C] Epithelial tissue
[D] Connective Tissue

14. Following part of the olfactory receptor will respond to inhaled chemicals
 [A] Olfactory epithelium
 [B] Supporting Cells
 [C] Olfactory hairs
 [D] Basal cells

15. A microbe is introduced into an environment where its natural metabolism results in the detoxification or break down of hazardous chemicals or pollutants is called as
 [A] Biodistribution
 [B] Bioremediation
 [C] Bio-fixation
 [D] Biohazard

16. Acetylcholine is rapidly broken down by_____ enzyme.
 [A] Acetylcholinesterase
 [B] Anti-cholinesterase
 [C] Cholinekinase
 [D] Hydrolyase

17. Incubation is a process which _____of microbes.
 [A] Inhibits the growth
 [B] Promotes the growth
 [C] Maintain the growth
 [D] None of the above

18. _____ is used in gram staining for decolourization of the microbes.
 [A] Crystal Violet
 [B] Iodine
 [C] Alcohol
 [D] Nigrosin

19. The Bacteriologist is the person who is involved in the study of
 [A] Fungi
 [B] Bacteria
 [C] Virus
 [D] Protozoa

20. The function of cell wall is to
 [A] Support
 [B] Maintain Shape
 [C] Support and Maintain Shape
 [D] Flexibility

SECTION – II

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| Q 2 | Attempt any <u>TWO</u> of the following | |
| A | Describe the anatomy of taste buds with the diagram. | 05 |
| B | State various functions of connective tissue. | 05 |
| C | What is the significance of skin as the sense organ? | 05 |

- Q 3** Attempt any **TWO** of the following
- A** Simplify the physiology of hearing with well labelled diagram of the ear. **05**
- B** Infer the principle of staining of the microbes and explain any one method in brief. **05**
- C** Recognize the applications of micro-organisms in pharmaceutical Industry. **05**

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Describe the Physiology of muscle contraction in brief. **05**
- B** Write a detailed note on Passive transport system across cell membrane. **05**
- C** Explain the structure of skeletal muscle in detail with diagram. **05**
- D** Define following terms: **05**
- a. Trippe
 - b. Fatigue
 - c. Myalgia
 - d. Muscular atrophy
 - e. Myositis
- E** Explain the functions of mitochondria and golgi bodies. **05**
- F** Describe the various parts of the eye with their physiology in brief. **05**
- Q 5** Attempt any **FOUR** of the following
- A** Draw a well labeled diagram of cell membrane. **05**
- B** Discuss in detail various properties of skeletal muscle. **05**
- C** Define cell division. Discuss mitosis in detail. **05**
- D** Explain Isolation and Incubation of micro-organism in brief. **05**
- E** Differentiate between eukaryotic and prokaryotic cell structures. **05**
- F** Write a note on active transport and vesicle mediated transport. **05**
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